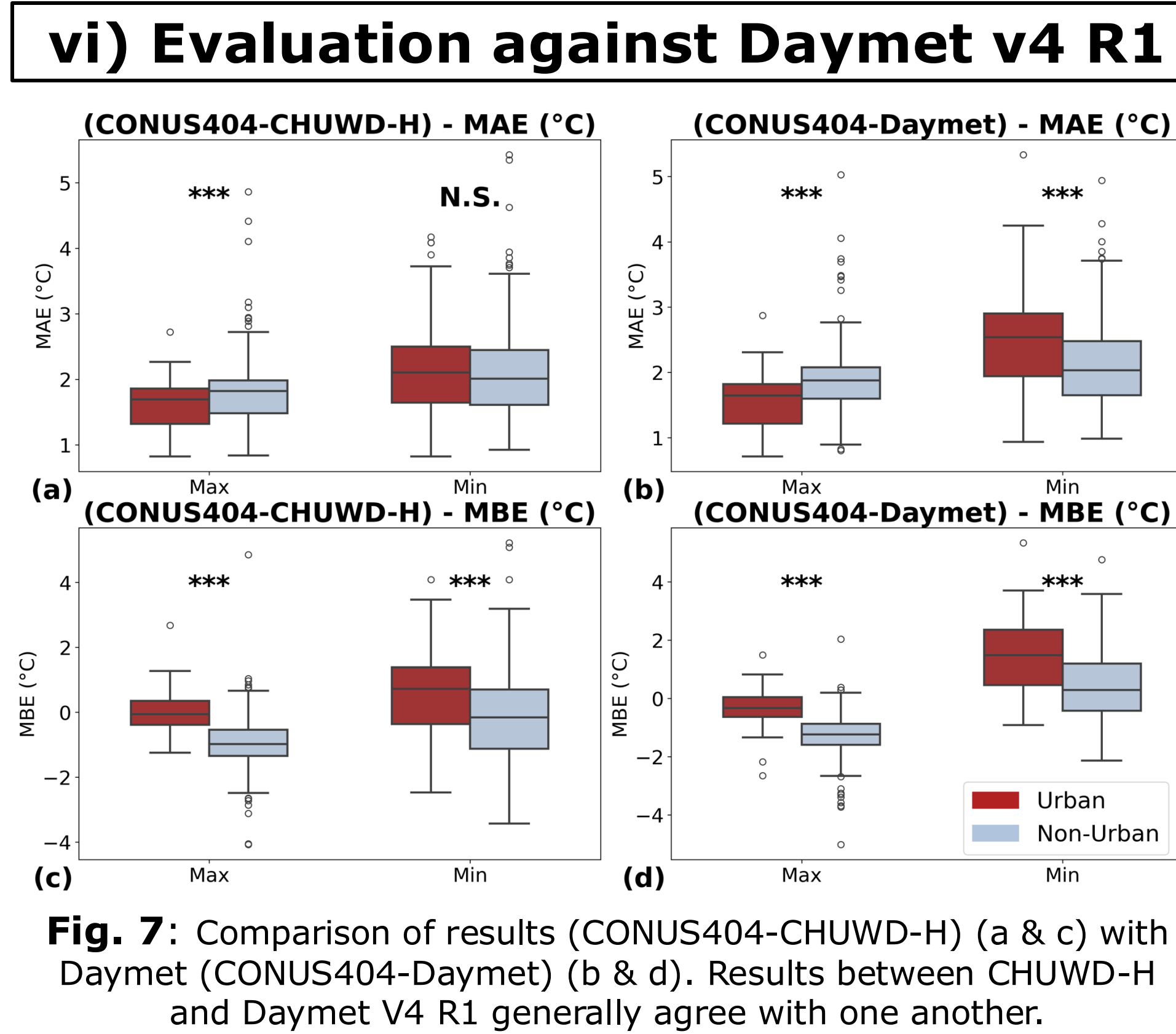
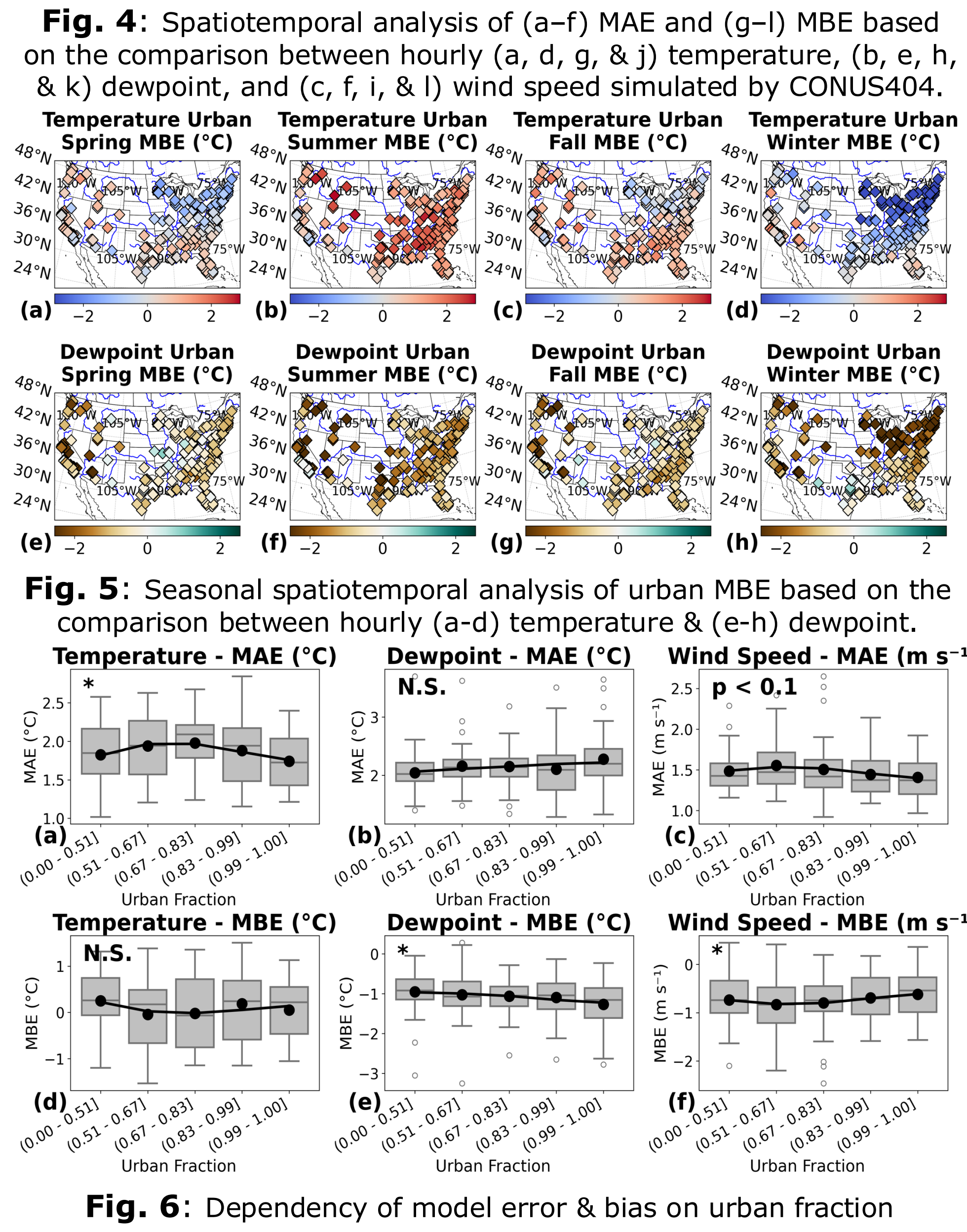
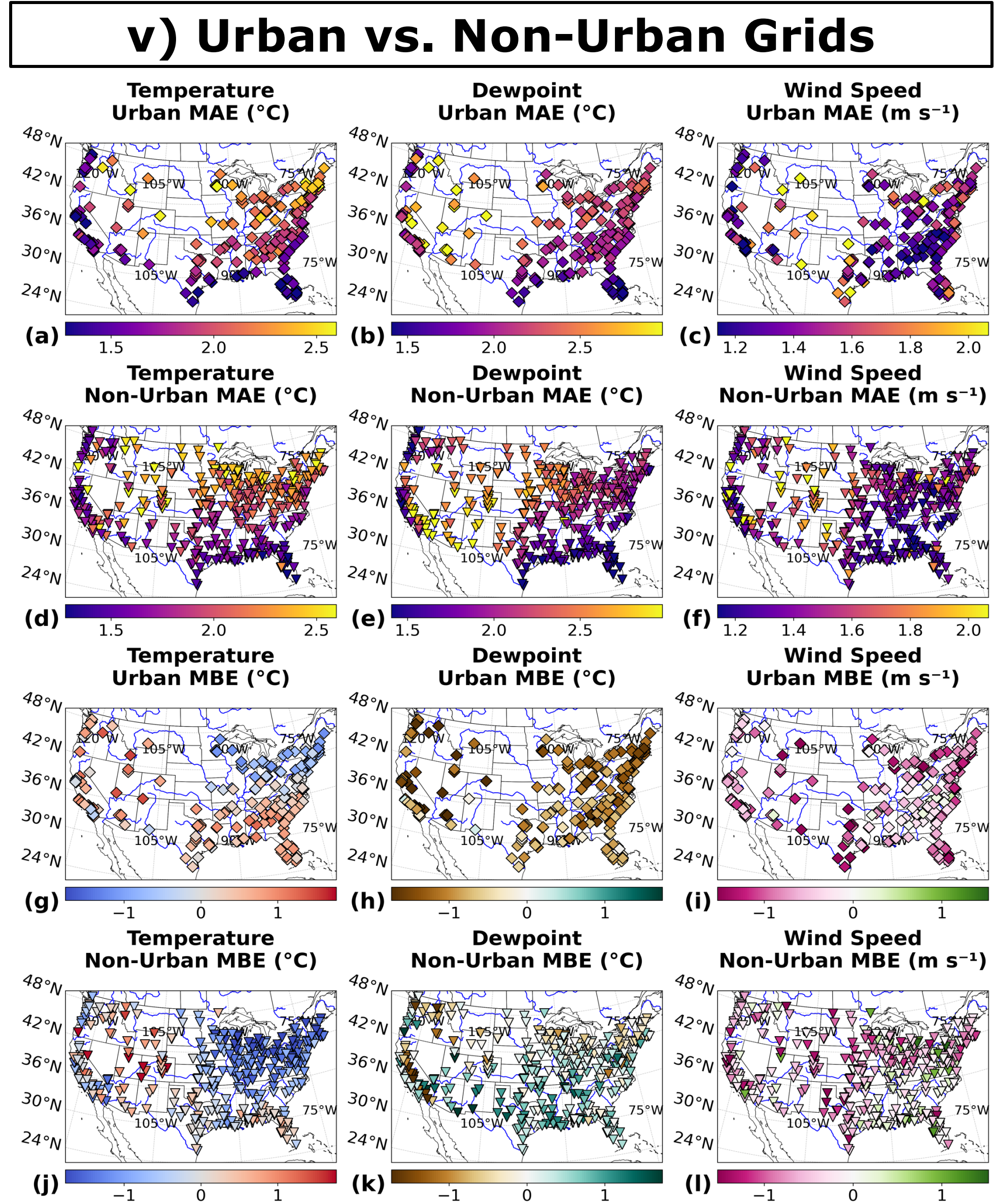
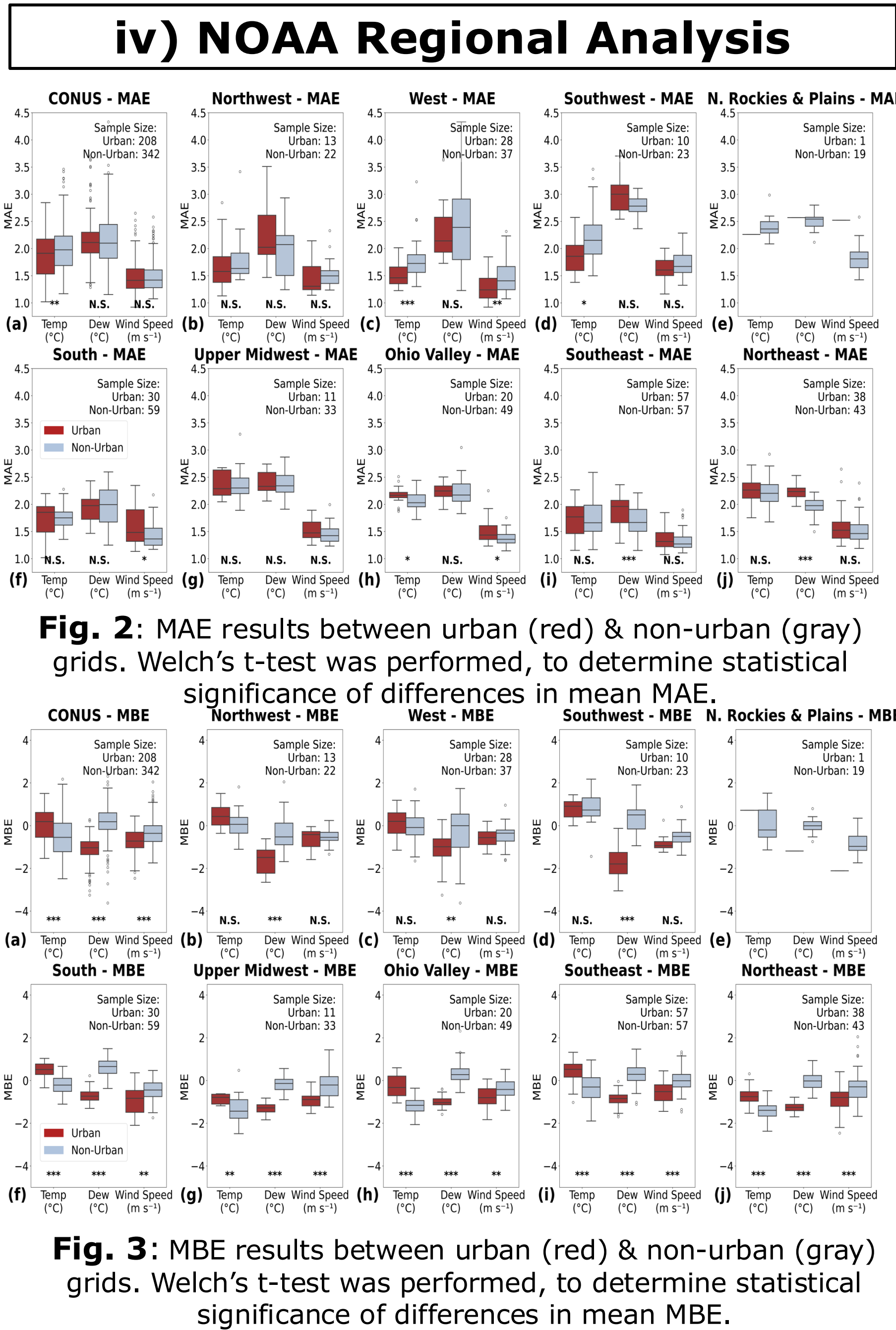
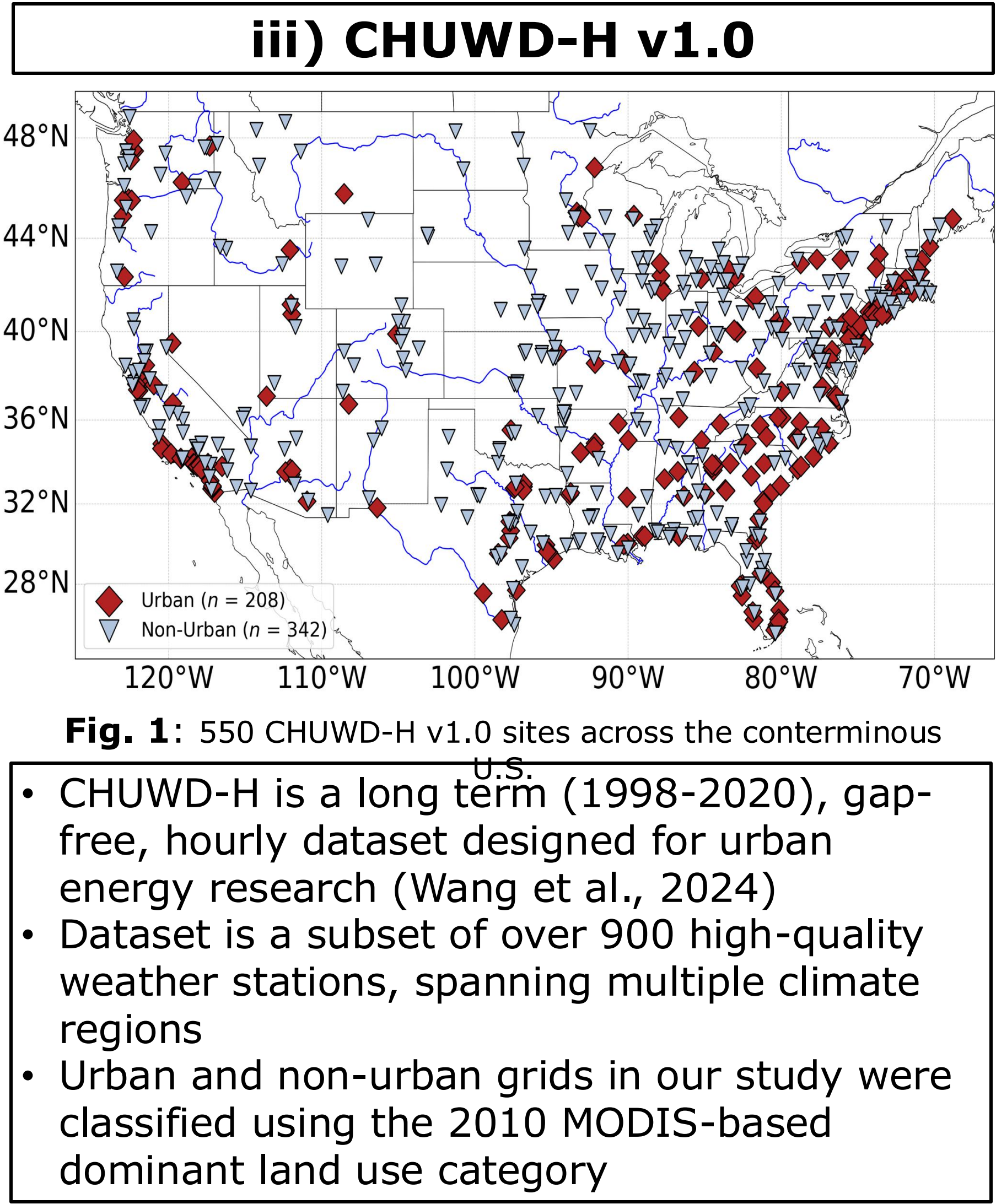


i) Background

- CONUS404 is one of the highest resolution (4-km) convection-permitting reanalysis datasets (Rasmussen et al., 2023), which could allow the model to be a critical tool in understanding unique urban dynamics.
- Accurate representations of urban features in numerical weather models are crucial for accurately simulating complex interactions between cities and the atmosphere.
- Accuracy of the bulk approach used in CONUS404 to simulate atmospheric conditions in urban areas at continental- and regional-scales remains unclear.
- Past studies have lacked a direct comparison of model errors & biases in urban grids to non-urban grids (Thompson et al., under review).
- These results have implications for improving the accuracy of simulated urban hydrometeorological conditions at different scales.

ii) Methodology

- Utilized the gap-filled Comprehensive Hourly Urban Weather Database (CHUWD-H v1.0) to derive hourly station-level, observed data
- Retrieved 10-years of CONUS404 2-m temperature, 2-m dewpoint, 10-m wind components (u & v), at 550 sites from the National Center for Atmospheric Research Research Data Archive
- Calculated mean absolute error (MAE)(tests accuracy) and mean bias error (MBE)(systematic bias) as CONUS404 – CHUWDH
- Applied and mapped seasonal analyses to gauge temporal performance
- Quantified the regionality of the model's errors and biases using the nine NOAA Climate Regions (Karl and Koss, 1984)
- Tested the dependency of model error on the urban fraction of the model grid
- Verified results from (CONUS404-CHUWDH) with an independent dataset



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